

DecodeLabs Data Cleaning & Exploratory Data Analysis Report

Project Overview

This project focused on performing data cleaning, validation, and exploratory data analysis (EDA) on a transactional business dataset using Python and Pandas.

The primary objective was to transform raw transactional data into a structured and analysis-ready dataset by identifying data quality issues, validating business logic, and ensuring dataset consistency and integrity.

The project simulated a real-world business analytics workflow commonly performed by data analysts before reporting and decision-making processes.

Tools & Technologies Used

- Python
 - Pandas
 - NumPy
 - Jupyter Notebook
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Dataset Overview

The dataset contained approximately 1,200 transactional records and 14 business-related variables, including:

- Order Information
- Customer IDs
- Product Details
- Payment Methods
- Quantity Purchased
- Unit Pricing
- Total Pricing
- Coupon Usage
- Transaction Dates

The dataset represented customer transaction activity and was analyzed to ensure data quality, reliability, and consistency for downstream analysis.

Data Cleaning & Validation Process

1. Missing Value Handling

The dataset was inspected for incomplete records and missing values across all variables.

Findings

- The `CouponCode` column contained missing values because not all customers applied discount coupons during transactions.

Action Taken

- Missing coupon entries were replaced with "No Coupon" to preserve dataset completeness while maintaining business context.

Business Justification

Replacing null values with a meaningful category prevented unnecessary data loss and improved interpretability during future analysis.

2. Duplicate Record Validation

Duplicate checks were performed to ensure transaction uniqueness and prevent analytical distortion.

Validation Checks

- Duplicate rows
- Duplicate `OrderID` values

Findings

- No duplicate records were identified.

Business Impact

This confirmed strong dataset integrity and ensured revenue calculations and transaction counts remained accurate.

3. Date Standardization & Validation

The `Date` column was validated to confirm consistency and proper datetime formatting.

Action Taken

- Verified successful datetime formatting across all records.

Business Importance

Consistent date formatting is essential for:

- time-series analysis,
 - trend analysis,
 - reporting accuracy,
 - and business intelligence workflows.
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4. Invalid Value Detection

Validation checks were conducted to identify impossible or suspicious transactional values.

Columns Validated

- `Quantity`
- `UnitPrice`
- `TotalPrice`

Findings

- No negative values were identified in transactional fields.

Business Importance

This confirmed that the dataset did not contain invalid sales transactions or corrupted financial records.

5. Financial Validation

Revenue integrity validation was performed to confirm that:

$\text{Quantity} \times \text{UnitPrice} = \text{TotalPrice}$

Findings

Initial inconsistencies were observed due to floating-point precision behavior in Python.

Action Taken

- Values were rounded to two decimal places before validation.

Result

All financial calculations were successfully validated after standardization.

Business Importance

Financial validation ensures reliable revenue reporting and prevents analytical inaccuracies in business reporting systems.

6. Text Standardization

Text-based columns were standardized to improve dataset consistency and readability.

Action Taken

- Applied consistent title formatting to categorical text fields.

Business Importance

Standardized text formatting improves:

- filtering consistency,
- grouping accuracy,
- reporting readability,
- and dashboard reliability.

Exploratory Data Analysis (EDA)

Initial exploratory analysis was conducted to understand dataset structure and transaction behavior.

The analysis included:

- summary statistics,
- transaction distribution review,
- missing value analysis,
- and financial validation checks.

Key metrics such as transaction counts, pricing consistency, and customer activity were explored to ensure the dataset was suitable for business analysis.

Key Project Outcomes

- Successfully transformed raw transactional data into a structured and analysis-ready dataset.
 - Applied professional data validation techniques to ensure dataset integrity.
 - Improved data consistency through text and datetime standardization.
 - Verified financial accuracy through revenue validation checks.
 - Developed practical experience in real-world data cleaning workflows using Python and Pandas.
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Conclusion

This project demonstrated the critical role of data cleaning and validation in preparing reliable datasets for analytical decision-making.

Through structured cleaning processes, validation checks, and exploratory analysis, the dataset was transformed into a high-quality analytical asset suitable for downstream business intelligence and reporting workflows.

The project also strengthened practical skills in:

- data preparation,
- business-rule validation,
- exploratory data analysis,
- and professional analytical documentation using Python and Pandas.